

Problem 2.2**Solution:**

- (a) (i) Subproblems:

For each $i = 1, 2, \dots, n$, define,

$$P(i) = \text{minimum penalty for subsequence } \langle a_i, \dots, a_n \rangle.$$

Want $P(0)$.

- (ii) Recursive formula:

$$P(i) = \min_{\substack{j \in (i+1, \dots, n) \\ i_j - i_{j-1} \leq 10 \text{ for all } j=1, \dots, k-1 \\ i_k - i_{k-1} \leq 5 \\ \alpha < L}} (P(j) + f(L - \alpha)),$$

where $\alpha = \sum_{m=i+1}^j a_m$. Base case is $P(n+1) = 0$.

- (iii) Evaluation order:

Pre-compute $S(i) = \sum_{m=i+1}^n a_m$, then evaluate $P(i)$ from n to 1.

- (iv) Time and space analysis:

Iteration takes $O(n)$ steps, each step evaluates $P(j)$ and $f(L - \alpha)$ in $O(1)$ time with pre-computed $S(i)$, total time complexity is $O(n)$. $P(i)$ and $S(i)$ both take $O(n)$ space, total space complexity is $O(n)$.

- (b) (i) Subproblems:

For each $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$, define,

$$P(i, j) = \text{minimum penalty for the } i\text{th subsequence ending at } a_j.$$

Want $P(m, n)$.

- (ii) Recursive formula:

$$P(i, j) = \min_{\substack{u \in (1, 2, \dots, j) \\ \alpha < L}} (P(i-1, u) + f(L - \alpha)),$$

where $\alpha = \sum_{m=u+1}^j a_m$. Base case is $P(0, 0) = 0$, $P(0, j) = +\infty$ for $j = 1, 2, \dots, n$.

- (iii) Evaluation order:

Pre-compute $S(i) = \sum_{v=i+1}^n a_v$, then evaluate i from 1 to m , with each i , evaluate j from 1 to n .

- (iv) Time and space analysis:

Iteration takes $O(mn)$ steps, each step evaluates minimum for $O(n)$ times, each evaluation takes $O(1)$ time with pre-computed $S(i)$, total time complexity is $O(mn^2)$. $P(i, j)$ takes $O(mn)$ space (or $O(n)$ space by using rolling array), $S(i)$ takes $O(n)$ space, total space complexity is $O(n)$.